

Selected Problems on Array & String

Try to solve the following problems without using any built-in functions in Java

1. Write a Java function to count the occurrence of a certain char in a string.

int countChar(String str, char c)

Input: str = "abcdefaa" c = 'a'

Output = 3

```
import java.util.Scanner;
public class Main {
    static int countChar(String str, char c) {
        int counter = 0;
        for (int i = 0; i < str.length(); i++) {
            if (str.charAt(i) == c) {
                counter+=1;
            }
        }
        return counter;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str = myObj.nextLine();
        char c = myObj.next().charAt(0);
        int result = countChar(str,c);
        System.out.println(result);
    }
}
```

Selected Problems on

Array & String

2. Write a Java function to replace only the first occurrence of a char in a string with a new char.

String replaceFirst(String str, char oldChar, char newChar)

Input: str = "abcdefaa" oldChar = 'a' newChar = 'H'

Output = "Hbcdefaa"

```
import java.util.Scanner;
public class Main {
    static String replaceFirst(String str, char oldChar, char newChar) {
        if (str == null || str.isEmpty()) {
            return str;
        }
        char[] charArray = str.toCharArray();
        for (int i = 0; i < charArray.length; i++) {
            if (charArray[i] == oldChar) {
                charArray[i] = newChar;
                break;
            }
        }
        return new String(charArray);
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str = myObj.nextLine();
        char c1 = myObj.next().charAt(0);
        char c2 = myObj.next().charAt(0);
        String result = replaceFirst(str, c1, c2);
        System.out.println(result);
    }
}
```

Selected Problems on

Array & String

3. Write a Java function to replace only the last occurrence of a char in a string with a new char.

String replaceLast(String str, char oldChar, char newChar)

Input: str = "abcdefaa" oldChar = 'a' newChar = 'H'

Output = "abcdefaH"

```
import java.util.Scanner;
public class Main {
    static String replaceLast(String str, char oldChar, char newChar) {
        if (str == null || str.isEmpty()) {
            return str;
        }
        char[] charArray = str.toCharArray();
        int lastIndex = -1;
        for (int i = 0; i < charArray.length; i++) {
            if (charArray[i] == oldChar) {
                lastIndex = i;
            }
        }
        if (lastIndex != -1) {
            charArray[lastIndex] = newChar;
        }
        return new String(charArray);
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str = myObj.nextLine();
        char c1 = myObj.next().charAt(0);
        char c2 = myObj.next().charAt(0);
        String result = replaceLast(str, c1, c2);
        System.out.println(result);
    }
}
```

Selected Problems on

Array & String

4. Write a Java function to replace each certain char in a string with a new char.

String replaceAll(String str, char oldChar, char newChar)

Input: str = "abcdefaa" oldChar = 'a' newChar = 'H'

Output = "HbcdefHH"

```
import java.util.Scanner;
public class Main {
    static String replaceAll(String str, char oldChar, char newChar) {
        if (str == null || str.isEmpty()) {
            return str;
        }
        char[] charArray = str.toCharArray();
        for (int i = 0; i < charArray.length; i++) {
            if (charArray[i] == oldChar) {
                charArray[i] = newChar;
            }
        }
        return new String(charArray);
    }

    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str = myObj.nextLine();
        char c1 = myObj.next().charAt(0);
        char c2 = myObj.next().charAt(0);
        String result = replaceAll(str, c1, c2);
        System.out.println(result);
    }
}
```

Selected Problems on

Array & String

5. Write a Java function to replace the occurrence of a char only at odd index in a string with a new char.

String replaceOddIndex(String str, char oldChar, char newChar)

Input: str = "abcdefaaderagg" oldChar = 'a' newChar = 'H'

Output = "abcdefaHderHgg"

```
import java.util.Scanner;
public class Main {
    static String replaceodd(String str, char oldChar, char newChar) {
        if (str == null || str.isEmpty()) {
            return str;
        }
        char[] charArray = str.toCharArray();
        for (int i = 0; i < charArray.length; i++) {
            if (i % 2 != 0) {
                if (charArray[i] == oldChar) {
                    charArray[i] = newChar;
                }
            }
        }
        return new String(charArray);
    }

    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str = myObj.nextLine();
        char c1 = myObj.next().charAt(0);
        char c2 = myObj.next().charAt(0);
        String result = replaceodd(str, c1, c2);
        System.out.println(result);
    }
}
```

Selected Problems on Array & String

6. Write a Java function to replace the occurrence of a char only at even index in a string with a new char.

String replaceEvenIndex(String str, char oldChar, char newChar)

Input: str = "abcdefaaderagg" oldChar = 'a' newChar = 'H'

Output = "abcdefHaderagg"

```
import java.util.Scanner;
public class Main {
    static String replaceeven(String str, char oldChar, char newChar) {
        if (str == null || str.isEmpty()) {
            return str;
        }
        char[] charArray = str.toCharArray();
        for (int i = 0; i < charArray.length; i++) {
            if (i % 2 == 0) {
                if (charArray[i] == oldChar) {
                    charArray[i] = newChar;
                }
            }
        }
        return new String(charArray);
    }

    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str = myObj.nextLine();
        char c1 = myObj.next().charAt(0);
        char c2 = myObj.next().charAt(0);
        String result = replaceeven(str, c1, c2);
        System.out.println(result);
    }
}
```

Selected Problems on Array & String

7. Write a Java function to concatenate a string to another.

String Concat(String S1, String S2)

Input: S1 = "Ahmed" S2 = "Ali"

Output = "Ahmed Ali"

```
import java.util.Scanner;
public class Main {
    static String Concat(String S1, String S2) {
        return S1 + S2;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        String str2 = myObj.nextLine();
        String result = Concat(str1, str2);
        System.out.println(result);
    }
}
```

8. Write a Java function to concatenate a string to another but in a reversed order

String ConcatReverse(String S1, String S2)

Input: S1 = "Ahmed" S2 = "Ali"

Output = "Ahmed ilA"

```
import java.util.Scanner;
public class Main {
    static String Concaterev(String S1, String S2) {
        char[] reversedS1 = new char[S2.length()];
        for (int i = S2.length() - 1, j = 0; i >= 0; i--, j++) {
            reversedS1[j] = S2.charAt(i);
        }
        return S1 + " " + new String(reversedS1);
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        String str2 = myObj.nextLine();
        String result = Concaterev(str1, str2);
        System.out.println(result);
    }
}
```

Selected Problems on Array & String

9. Write a Java function to concatenate a string to another but in a special way that append the second string in the odd indices only.

String ConcatSpecial(String S1, String S2)

Input: S1 = “Ahmed” S2 = “Ali”

Output = “AAhlmied”

```
import java.util.Scanner;
public class Main {
    static String ConcatSpecial(String S1, String S2) {
        char[] result = new char[S1.length() + S2.length()];
        int i = 0, j = 0;
        for (int k = 0; k < result.length; k++) {
            if (k % 2 == 0 && i < S1.length()) {
                result[k] = S1.charAt(i);
                i++;
            } else if (j < S2.length()) {
                result[k] = S2.charAt(j);
                j++;
            }
        }
        return new String(result);
    }

    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        String str2 = myObj.nextLine();
        String result = ConcatSpecial(str1, str2);
        System.out.println(result);
    }
}
```


Selected Problems on Array & String

10. Write a Java function to reverse a string.

String reverseString(String S)

Input: S = "Ahmed"

Output = "demhA"

```
import java.util.Scanner;
public class Main {
    static String reverseString(String S1) {
        String reverse = "";
        for (int i = S1.length() - 1, j = 0; i >= 0; i--, j++) {
            reverse += S1.charAt(i);
        }
        return new String(reverse);
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        //String str2 = myObj.nextLine();
        String result = reverseString(str1);
        System.out.println(result);
    }
}
```

Selected Problems on Array & String

11. Write a Java function to reverse only the first word in a string.

String reverseFirstWord(String S)

Input: S = “Ahmed Ali”

Output = “demhA Ali”

```
import java.util.Scanner;
public class Main {
    static void reverseFirstWord(String s) {
        int space = 0;
        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == ' ') {
                space = i;
                break;
            }
        }
        for (int i = space - 1; i >= 0; i--) {
            System.out.print(s.charAt(i));
        }
        for (int i = space ; i < s.length(); i++) {
            System.out.print(s.charAt(i));
        }
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        //String str2 = myObj.nextLine();
        reverseFirstWord(str1);
    }
}
```

Selected Problems on

Array & String

12. Write a Java function to reverse only the last word in a string.

String reverseLastWord(String S)

Input: S = “Ahmed Mohamed Ali”

Output = “Ahmed Mohamed ilA”

```
import java.util.Scanner;
public class Main {
    static String reverseLastWord(String s) {
        int space = 0;
        String before_last_space = "", after_last_space="";
        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == ' ') {
                space = i;
            }
        }
        for (int i = 0; i < space; i++) {
            before_last_space += s.charAt(i);
        }
        for (int i = s.length()-1; i > space; i--) {
            after_last_space += s.charAt(i);
        }
        return before_last_space + " " + after_last_space;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        //String str2 = myObj.nextLine();
        System.out.println(reverseLastWord(str1));
    }
}
```

Selected Problems on Array & String

13. Write a Java function to convert a string to array of char.

Char[] stringToCharArray(String S)

Input: S = "Ahmed"

Output = {'A', 'h', 'm', 'e', 'd'}

```
import java.util.Scanner;
public class Main {
    static char[] stringToCharArray(String s) {
        char[] charArray = new char[s.length()];
        for (int i = 0; i < s.length(); i++) {
            charArray[i] = s.charAt(i);
        }
        return charArray;
    }

    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String str1 = myObj.nextLine();
        //String str2 = myObj.nextLine();
        char[] arr = new char[str1.length()];
        arr = stringToCharArray(str1);
        System.out.println(arr);
    }
}
```

Selected Problems on Array & String

14. Write a Java function to convert array of char to a string.

String charArrayToString(char[] c)

Input: c = {'A', 'h', 'm', 'e', 'd'}

Output = "Ahmed"

```
import java.util.Scanner;
public class Main {
    static String charArrayToString(char[] a){
        String s="";
        for (int i = 0; i < a.length; i++) {
            s += a[i];
        }
        return s;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        //String str1 = myObj.nextLine();
        //String str2 = myObj.nextLine();
        char[] arr = {'a' , 'l' , 'i'};
        String s = charArrayToString(arr);
        System.out.println(s);
    }
}
```

Selected Problems on

Array & String

15. Write a Java function to split a string with a specific char and put the result into array of String.

String [] Split (String S, char c)

Input: S = "Ahmed Mohamed Ali Mousa"

Output: words = {"Ahmed", "Mohamed", "Ali", "Mousa"}

```
import java.util.Scanner;
public class Main {
public static String[] split(String s, char c) {
    int count = 1;
    for (int i = 0; i < s.length(); i++) { if (s.charAt(i) == c) {count++; } }
    String[] words = new String[count];
    int startIndex = 0, arrayIndex = 0;
    for (int i = 0; i < s.length(); i++) {
        if (s.charAt(i) == c) {
            words[arrayIndex] = Substring_notbuilt(s, startIndex, i);
            startIndex = i + 1;
            arrayIndex++;
        }
    }
    words[arrayIndex] = Substring_notbuilt(s, startIndex, s.length());
    return words;
}

private static String Substring_notbuilt(String s, int start, int end) {
    /*private it means that it is only accessible within the same class in
which it is declared*/
    if (start >= end) {
        return "";
    }
    char[] result = new char[end - start];
    int resultIndex = 0;
    for (int i = start; i < end; i++) {
        result[resultIndex] = s.charAt(i);
        resultIndex++;
    }
    return new String(result);
}

public static void main(String[] args) {
    Scanner myObj = new Scanner(System.in);
    String s1 = myObj.nextLine();
    char c = ' ';
    String[] result = (split(s1,c));
    for (int i = 0; i < result.length; i++) {
        System.out.print(result[i]);
        System.out.print(" ");
    }
}
```

Selected Problems on Array & String

16. Write a Java function to check if a string start with a specific substring or not.

boolean startWith (String S, String sub)

Input: S = "abcd" sub = "cd"

Output: No

Input: S = "abcd" sub = "ab"

Output: Yes

```
import java.util.Scanner;
public class Main {
    static boolean startWith(String s, String sub){
        boolean start=true; int index=0;
        for(int i=0;i<sub.length();i++){
            if(sub.charAt(i)!=s.charAt(index)) start=false;
            index++;
        }
        return start;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        String s2 = myObj.nextLine();
        System.out.println( (startWith(s1, s2)) ? "Yes" : "No" );
    }
}
```

Selected Problems on

Array & String

17. Write a Java function to check if a string end with a specific substring or not.

boolean endWith (String S, String sub)

Input: S = "abcd" sub = "cd"

Output: Yes

Input: S = "abcd" sub = "ab"

Output: No

```
import java.util.Scanner;
public class Main {
    static boolean endWith(String s, String sub){
        boolean end=true; int index=0;
        for(int i=sub.length();i<s.length();i++){
            if(s.charAt(i)!=sub.charAt(index)) start=false;
            index++;
        }
        return end;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        String s2 = myObj.nextLine();
        System.out.println( (endWith(s1, s2)) ? "Yes" : "No" );
    }
}
```


Selected Problems on Array & String

18. Write a Java function to find the index within a string of the last occurrence of the specified char.

int lastIndex (String S, char c)

Input: S = “Ahmed Mohamed Ali” char = ‘d’

Output: index = 12

```
import java.util.Scanner;
public class Main {
    static int lastIndex (String S, char c){
        int index = 0;
        for (int i = 0; i < S.length(); i++) {
            if (S.charAt(i) == c) {
                index =i;
            }
        }
        return index;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        char c = myObj.next().charAt(0);
        System.out.println( (lastIndex(s1, c)));
    }
}
```

Selected Problems on

Array & String

19. Write a Java function that remove the leading and trailing spaces in a string.

String trimString (String S)

Input: S = “ Ahmed Mohamed Ali ”

Output: S = “Ahmed Mohamed Ali”

```
import java.util.Scanner;
public class Main {
    static String trimString (String s){
        String news="";
        for(int i=0;i<s.length();i++){
            news+= (s.charAt(i) == ' ' && (i==0 || i==s.length()-1)) ? " " :
s.charAt(i);
        }
        return news;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        System.out.println( (trimString (s1)));
    }
}
```

Selected Problems on

Array & String

20. Write a Java function that remove a certain char from a string.

String removeChar (String S, char c)

Input: S = "Ahmed Mohamed Ali" char = 'm'

Output: S = "Ahed Mohaed Ali"

```
import java.util.Scanner;
public class Main {
    static String removeChar (String S, char c){
        String result = "";
        for (int i = 0; i < S.length(); i++) {
            result += (S.charAt(i) == c) ? "" : S.charAt(i);
        }
        return result ;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        char c = myObj.next().charAt(0);
        System.out.println( (removeChar (s1, c)));
    }
}
```

21. Write a Java function that duplicate all chars in a string.

String duplicateAllChar (String S)

Input: S = "Ahmed"

Output: S = "AAhhmmeedd"

```
import java.util.Scanner;
public class Main {
    static String duplicateAllChar (String S){
        String result = "";
        for (int i = 0; i < S.length(); i++) {
            result += S.charAt(i);
            result += S.charAt(i);
        }
        return result ;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        System.out.println( (duplicateAllChar (s1)));
    }
}
```

Selected Problems on Array & String

22. Write a Java function that duplicate a certain char in a string.

String duplicateChar (String S, char c)

Input: S = "Ahmed" char = 'm'

Output: S = "Ahmmed"

```
import java.util.Scanner;
public class Main {
    static String duplicateChar (String S, char c){
        String result = "";
        for (int i = 0; i < S.length(); i++) {
            result += S.charAt(i);
            if ( S.charAt(i) == c) {
                result += S.charAt(i);
            }
        }
        return result ;
    }
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String s1 = myObj.nextLine();
        char c = myObj.next().charAt(0);
        System.out.println( (duplicateChar (s1,c)));
    }
}
```